Charlie Mai

Cloud, DevOps & Production Reliability Engineer

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PROFILE

Cloud, DevOps and production-reliability engineer with 17+ years across software delivery, distributed systems, technical leadership and production operations. Career since Feb 2009. Former AWS Cloud Support Engineer II and CloudFront SME with deep experience in incident investigation, root-cause analysis and customer-facing technical resolution. Currently operating AWS-based systems and building practical automation for complex integration and support workflows. Applies AI selectively to evidence-grounded troubleshooting and operational decision support.

CORE CAPABILITIES

Production reliability Incident response, RCA, observability, failure analysis	**Cloud and distributed systems** AWS, Lambda, Step Functions, EventBridge, CloudFront, S3, Route 53, IAM, WAF
DevOps and automation Python, CI/CD, test automation, release support, infrastructure automation	**Integration and backend** REST APIs, PostgreSQL, event-driven architecture, diagnostic tooling
Applied AI RAG, retrieval evaluation, evidence-grounded assistants, human-in-the-loop design	**Target opportunities** Cloud, DevOps, Production Reliability, Platform Engineering and SRE-aligned roles in Ireland and the EU

SELECTED PROOF

- **17+** - years across engineering and operations; career since Feb 2009
- **440k+** - civic reports handled by a public platform
- **#1** - customer satisfaction ranking within an AWS support team
- **457** - AWS support cases resolved; Top 2 case-performance ranking

EXPERIENCE

IT DevOps Support Engineer at National Broadband Ireland

Feb 2025 - Present - Dublin, Ireland

Operate and troubleshoot event-driven AWS systems and commercial platforms, with a focus on production reliability and actionable diagnostics.

- Operate event-driven AWS systems integrating Lambda, Step Functions, EventBridge, API integrations, PostgreSQL and commercial platforms.
- Investigate production incidents, integration failures and recurring defects using logs, data analysis and evidence-driven RCA.
- Build repeatable test, deployment and diagnostic workflows; develop Python automation and observability tooling.
- Support production releases, maintenance activities and cross-system workflows involving cloud services, internal platforms and third-party applications.

Business Programme Manager at VIVERSE

Apr 2024 - Feb 2025 - Dublin, Ireland

Led cross-functional delivery and customer-facing technical coordination for product and enterprise initiatives across EMEA.

- Managed parallel workstreams across engineering, R&D and business stakeholders; aligned scope, requirements and delivery plans.
- Connected technical teams and business owners so decisions moved from ambiguity to delivery.

Cloud Support Engineer II / Cloud Support Engineer at Amazon Web Services (AWS)

Jan 2022 - Mar 2024 - Dublin, Ireland

Progressed to Cloud Support Engineer II within 21 months and became a CloudFront SME while resolving complex distributed-cloud issues.

- Resolved 457 complex AWS support cases across CloudFront, S3, Route 53, API Gateway, Lambda, IAM, WAF, SES and media services.
- Ranked #1 for customer satisfaction and Top 2 for case performance within an AWS cloud support team.
- Accredited as a CloudFront SME; contributed to mentoring, enablement, documentation and hiring.
- Worked with customers, architects and TAMs to reproduce issues, provide root-cause analysis and support critical incident communications.

EXPERIENCE CONTINUED

Solutions Engineer / Delivery Customer Support Engineer at Verizon Media Platform

Apr 2019 - Aug 2021 - Taiwan

Delivered customer-facing CDN solutions, implementation support and incident coordination for enterprise media customers.

- Analysed traffic logs and CDN performance to support production troubleshooting and service optimisation.
- Managed incident, problem and change workflows with regional teams during onboarding and production issues.
- Supported enterprise implementations, PoCs and cross-regional delivery with engineering and commercial stakeholders.

Technical Lead at CATCHPLAY

May 2017 - Apr 2019 - Taipei / Jakarta

Led Android TV and set-top-box delivery, partner integrations and cross-platform technical alignment across Android, iOS and web teams.

- Built CI practices and helped improve application architecture, performance and user experience.
- Supported streaming SDK integration and multi-region product delivery across Taiwan and Indonesia.

Earlier Engineering & Technical Leadership

Team Lead, Mobile / Android Developer / IT Lead

2009 - 2017 - Taiwan

Built software products and led mobile, backend and application delivery across several technology companies.

- Led native mobile development, CI setup, technical capability building and application architecture improvements.
- Worked across Android, iOS, web, hybrid applications, Java enterprise systems and internal business platforms.

SELECTED PROJECTS

RAG Troubleshooting Assistant - Active lab

Python service prototype for document ingestion, semantic retrieval, source visibility and human review; evaluation contract covers missing context, ambiguity, abstention and grounded next actions. No production adoption claim.

Parking Violation Reporter - Public impact case study

Mobile workflow: capture violation details, attach evidence, submit a civic report and receive status or feedback; the platform handled more than 440,000 reports and the related concept received a U-START Silver Medal.

Vouchgether - Historical case study

Documented flow using React Native, Cognito, Lambda, S3 and DynamoDB across voucher discovery, sharing and claiming; historical version only, with no current live architecture claim.

OPERATING PRINCIPLES

- Make failure behaviour visible and the next engineering action explicit.
- Keep measured evidence, historical context and current status separate.
- Use human review when an automated answer could change a high-consequence decision.
- Prefer reproducible checks and small automation before adding platform complexity.

EDUCATION

MSc in Computer Science (Negotiated Learning)
University College Dublin - Sep 2025 - Aug 2027; part-time study with an AI-focused pathway; expected 2027.

Certificate in Artificial Intelligence
University of Limerick - Sep 2025 - Jan 2026; First Class Honours.

Master of Information Technology
Queensland University of Technology - 2007 - 2008; GPA 6.5 / 7.0; Dean's List and scholarship recognition.

Bachelor of Civil Engineering
National Chiao Tung University - 2003 - 2006.

CERTIFICATIONS AND RECOGNITION

- AWS Certified Solutions Architect - Associate
- U-START Social Enterprise competition - Silver Medal
- Earlier Java, Java EE and Android certifications
- Languages: English (Full professional); Mandarin (Native or bilingual)